MAD LAB QUESTIONS

Model Question Bank (Diverse Multi-Activity Architecture Patterns)

Exam Blueprint Note: Every question below assesses the identical syllabus knowledge base—Layouts, Input Validation, SharedPreferences, Options/Context Menus, ListView Adapters, Explicit/Implicit Intents, Alert Dialogs, and Toasts—but presents structurally distinct application interaction flows and logic workflows.

1. Custom Typography Studio

Design and develop an Android application across two activities. The first activity displays a **ListView** of available application text fonts and layout theme colors using an Android Adapter. Long-pressing a theme item must open a Context Menu; selecting "Apply Strategy" must save the choice in Shared Preferences and launch the second activity via an Explicit Intent. The second activity must read these preferences to dynamically style its background layout, provide **EditText** form widgets for a user to type a custom headline, and feature an Options Menu with a "Validate & Preview" action. Clicking this menu option must validate that the text fields are not empty, showing a Toast upon success or a targeted popup if validation fails.

2. Inventory Reorder Deck

Design and develop an Android application featuring an inventory view activity and an item request activity. The first activity acts as a primary dashboard, utilizing a structured layout to display current warehouse tracking metrics along with an Options Menu containing an "Add Stock Request" action. Clicking this action triggers an Explicit Intent to open the second activity. The second activity features form widgets (**EditText**) for Item Name and Quantity, enforcing strict runtime field validation. Upon tapping a "Submit" button, an Alert Dialog must prompt for confirmation; once confirmed, the data is bundled and sent back to the first activity as an activity result. The first activity catches this result, saves the item name to Shared Preferences, and shows a Toast confirming the dashboard update.

3. Real Estate Agent Connector

Design and develop a property matching Android application. The first activity must immediately present an array of property listings inside a **ListView** utilizing an Android Adapter. Clicking any property item must spawn a confirmation Alert Dialog. If the user accepts, an Explicit Intent must bundle the property metadata and pass it to the second activity. The second activity displays these details cleanly and features an **EditText** feedback form widget with mandatory character length validation. This activity must include a Context Menu tied to the property header text, featuring an option called "Contact Representative" that fires an Implicit Intent to open the external device telephone dialer with the broker's number pre-loaded.

4. Secure Corporate Vault Browser

Design and develop a secure document organizer application. The launcher activity must feature an **EditText** credential login layout requiring valid code patterns (enforced by runtime input validation) and an Options Menu item to quickly wipe out typed characters. Upon successful entry, the app must store a "Session Active" flag in Shared Preferences and transition to the second activity via an Explicit Intent. The second activity displays corporate folders inside an adapter-linked **ListView**. Clicking any folder displays a temporary Toast, while long-pressing a folder activates a Context Menu that triggers an Implicit Intent to launch an external web browser to download document guidelines.

5. Smart Travel Hub Link Handler

Design and develop an Android travel planner capable of managing shared travel URLs. The first activity must configure an intent filter within its **AndroidManifest.xml** to intercept text/plain geography URIs shared from external apps. When triggered via this Implicit Intent, the activity must extract the incoming link data, run a validation check to ensure it is a valid address format, and use an Options Menu option to explicitly forward this data to the second activity. The second activity must append the incoming destination to an active trip history array displayed inside a **ListView** via an Android Adapter. Long-pressing a trip logs it into Shared Preferences and triggers an Alert Dialog confirmation.

6. Freelance Project Bidding Board

Design and develop a freelance project bidding utility. The first activity must utilize a structured layout to display form widgets for capturing a developer's Name, Bid Amount, and Project Scope, incorporating strict numeric validation rules on the bid field. When the user selects a "Post Bid" option from the activity's Options Menu, the application must store the Developer's Name using Shared Preferences, bundle the remaining metrics, and navigate to the second activity via an Explicit Intent. The second activity must display a static greeting using the preference data and map related tech stacks inside an adapter-backed **ListView**. Long-pressing a stack item must show a Context Menu that opens an Implicit Intent routing directly to a programming documentation web link.

7. Gym Trainer Allocation Portal

Design and develop a fitness tracking application. The first activity presents a profile setup wizard featuring an **EditText** for Username and Target Weight, requiring non-empty input validation. Clicking a "Setup Complete" button triggers an Alert Dialog; upon confirmation, the Username is saved into Shared Preferences, and an Explicit Intent opens the second activity. The second activity reads the username from preferences to build a dynamic banner, and loads available personal trainers into a **ListView** using an Android Adapter. Clicking a trainer item fires a Toast alert, whereas long-pressing the trainer reveals a Context Menu that executes an Implicit Intent to open the system email client to schedule a consultation session.

8. Vehicle Service Scheduler

Design and develop a vehicle service application. The first activity displays current vehicle diagnostics using structured view elements. The action flow relies on an Options Menu; selecting "Schedule Diagnostic" prompts an Alert Dialog, which redirects the user to the second activity using an Explicit Intent. The second activity consists of a detailed text layout and a validation-controlled **EditText** field to record specific mechanical issues. The logged issues are systematically structured into an adapter-driven **ListView**. Long-pressing any logged issue item generates a Context Menu that fires an Implicit Intent to launch an external map pinpointing the nearest authorized service center.

9. E-Banking Payee Hub

Design and develop an electronic banking transaction manager. The first activity captures a user's Bank Account ID using validation-checked **EditText** boxes and an Options Menu to clear incorrect numbers. Successfully saving this ID into Shared Preferences immediately fires an Explicit Intent opening the second activity. The second activity dynamically references the stored preferences and renders a list of registered payees inside a **ListView** via an Android Adapter. Clicking a payee item triggers an Alert Dialog demanding a transaction amount entry. Entering an amount and pressing confirm triggers a safety validation check, issuing a successful completion Toast and launching an Implicit Intent browser transaction link.

10. Digital Library & Book Checkout

Design and develop a library management application. The first activity populates a collection of academic textbooks inside a scrollable **ListView** using an Android Adapter. Clicking a book item displays a summary Toast, while long-pressing it populates a Context Menu. Selecting "Borrow Book" bundles the book's ISBN code and triggers an Explicit Intent to open the second activity. The second activity displays the book data, contains input fields for Borrower Name and Return Duration (complete with data validation check loops), and features an Options Menu with a "Finalize Check out" choice that updates Shared Preferences before executing an Implicit Intent to calendar applications.